

## Lab 08 - Expected verification evidence

Run from the lab directory. Run it after the failure injection (section 6) but before the fix, so the `BackoffLimitExceeded` assertions are exercised:

```
bash verification/checks.sh
```

### Expected transcript (single-node kind cluster)

```
Lab 08 verification - namespace kcna-lab08

== Namespace and datasets
[PASS] namespace kcna-lab08 is Active (= Active)
[PASS] datasets read: 8 berths (3 high-occupancy), 6 queue items, 2 tariff lanes under review
[PASS] ConfigMap berth-data carries all three data files

== DaemonSet berth-agent (one Pod per node)
[PASS] cluster has 1 schedulable node(s) - a DaemonSet must produce exactly that many Pods
[PASS] desiredNumberScheduled equals the node count (= 1)
[PASS] numberReady equals the node count (= 1)
[PASS] a DaemonSet has no .spec.replicas field (= )
[PASS] agent log reports the dataset it read (= berths=8 high_occupancy=3)

== Indexed Job manifest-reconcile
[PASS] completionMode (= Indexed)
[PASS] completions matches the queue length (= 6)
[PASS] parallelism (= 2)
[PASS] succeeded Pods (= 6)
[PASS] Job condition Complete (= True)
[PASS] one Pod per completion index (= 6)
[PASS] distinct work items reported RECONCILED (= 6)

== CronJob tariff-sync
[PASS] schedule (= */2 * * * *)
[PASS] concurrencyPolicy (= Forbid)
[PASS] successfulJobsHistoryLimit (= 3)
[PASS] CronJob has spawned 2 Job(s), within successfulJobsHistoryLimit=3
[PASS] a tariff-sync run printed SYNC-OK
[PASS] tariff-sync flagged the lanes marked 'review' in tariff-rates.csv (>=2 lines, found 2)

== Failure injection: customs-export exceeded its backoffLimit
[PASS] customs-export failure reason (= BackoffLimitExceeded)
[PASS] failed Pod attempts (1 initial + backoffLimit 2) (= 3)

-----
Lab 08: 23 passed, 0 failed
ALL CHECKS PASSED
```

```
echo $?
```

```
0
```

### Supporting evidence the learner should be able to produce

1. The DaemonSet count is derived from the node set - one node, one Pod

```
kubectl get nodes --no-headers | wc -l
kubectl -n kcna-lab08 get ds berth-agent \
  -o jsonpath='{desired={.status.desiredNumberScheduled} ready={.status.numberReady}}'
```

```
1
desired=1 ready=1
```

There is no .spec.replicas to print:

```
kubectl -n kcna-lab08 get ds berth-agent -o jsonpath='{.spec.replicas}'; echo "<-- empty"
```

```
<-- empty
```

2. The agent's reading of data/berth-readings.csv matches the file

```
kubectl -n kcna-lab08 logs -l app=berth-agent --tail=1
awk -F, 'NR>1 && $5+0 >= 90 {print $2, $5"%"}' data/berth-readings.csv
```

```
02:14:47 node=kind-control-plane berths=8 high_occupancy=3
B01 92%
B03 96%
B06 90%
```

3. Six indexes, six distinct work items, none repeated

```
kubectl -n kcna-lab08 get job manifest-reconcile \
  -o jsonpath='{completed={.status.completedIndexes} succeeded={.status.succeeded}}'
kubectl -n kcna-lab08 logs -l job-name=manifest-reconcile --tail=-1 | grep -c RECONCILED
```

```
completed=0-5 succeeded=6
6
```

4. The CronJob's reading of data/tariff-rates.csv

```
tariff-sync run at 2026-09-05T02:20:03Z
lanes=8 mean_rate_sgd=683.66
FLAGGED SG-ID Singapore to Tanjung Priok
FLAGGED SG-AE Singapore to Jebel Ali
SYNC-OK
```

5. The failure injection - three attempts, then the controller gives up

```
kubectl -n kcna-lab08 describe job customs-export | sed -n '/^Events:/, $p'
```

```
Events:
  Type     Reason             Age   From           Message
  ----     -
  Normal   SuccessfulCreate    2m10s job-controller Created pod: customs-export-9wq4k
  Normal   SuccessfulCreate    2m    job-controller Created pod: customs-export-cn2xr
  Normal   SuccessfulCreate    95s   job-controller Created pod: customs-export-4tbvz
  Warning   BackoffLimitExceeded 89s   job-controller Job has reached the specified backoff limit
```

with the application-level cause in the Pod log:

```
customs-export starting
wc: /data/customs-export.csv: No such file or directory
EXPORT-FAILED: source file missing
```

## Notes on variance

- \* Node count. The DaemonSet assertions compare against the live kubectl get nodes count, not a hard-coded 1. On a 3-node kind cluster the same script passes with desired=3.
- \* CronJob Job count. CronJob has spawned N Job(s) accepts 1-3 depending on how long the lab has been running and whether tariff-sync-manual was created. Manually created Jobs also carry app=tariff-sync and are counted.
- \* If you run checks.sh after repairing the customs-export Job, the last block prints the single line `"customs-export Job not present - failure injection is optional, section 6 skipped"` only when the Job has been deleted; if the repaired Job exists and is Complete, the Failed condition is absent and the same optional-skip line is printed. Total pass count then drops to 22.
- \* Pod name random suffixes, timestamps and AGE columns always differ.